

Leonardo Tonelli

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EDUCATION

EPFL - École Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

Title: MSc. in Statistics (minor in Data Science)

Sept 2025 - Present

GPA: 5.55/6.0

Coursework: Statistical Inference, Math of Data, Statistical Machine Learning, Deep Learning, Reinforcement Learning

BOCCONI UNIVERSITY

Milan, Italy

Title: BCs in Economics and Computer Science

Sept 2022 - July 2025

Final Mark: 110/110 Cum Laude

Thesis: "Replicated Approach to the Binary Perceptron" (supervised by Prof. Luca Saglietti)

Coursework: Advanced Statistics, Econometrics, Applied Mathematics, Machine Learning, Computer Programming

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Singapore,

Title: Exchange Student in Business & Computing programme

Jan 2025 - May 2025

90th percentile over 1179 exchange undergraduate winning applicants' scores.

Coursework: Business Analytics consulting (A+), Signal and Noise in Biology (A+) and Intelligent Agents (A+)

PROFESSIONAL EXPERIENCES

RESEARCH INTERN - Okinawa Institute of Science and Technology

Okinawa, Japan

- Currently doing research on Representation Learning under supervision of prof. Makoto Yamada. Focus on implementation of JEPA inspired architecture and investigate its workings under causal interventional settings.

Jul 2026 - Sep 2026

RESEARCH ASSISTANT - Bocconi University

Milan, Italy

- Analyzed RCT data (~1,200 participants; ~1M eye-tracking observations) to estimate treatment effects using ANOVA and regularized GLMs
- Built reproducible Python pipelines for large-scale behavioral data preprocessing and model validation
- Collaborated cross-functionally to translate statistical findings into policy-relevant insights

Sept 2024 - Dec 2024

SOFTWARE DEVELOPER - Obloo Ventures, Venture Capital and Private Equity

Milan, Italy

- Developed NLP application for audio transcription, summarization, and topic extraction.
- Implemented and tested RAG architecture; boosted efficiency by 70%, increased value extraction by 3x
- Technologies:** Docker, Streamlit, Groq, Hugging Face, Git

Jun - Sept
2024

SELECTED PROJECTS

Frame-Level Multimodal Hate Localization in Video

Built an end-to-end PyTorch pipeline for frame-level localization of hateful video content using visual, audio, speech, and OCR features from SigLIP2, WavLM, Whisper, HateBERT, and EasyOCR. Designed and evaluated coarse-to-fine multimodal architectures and ablations, achieving 0.733 frame mAP and 0.852 AUC on HateMM, outperforming the referenced MultiHateLoc baseline by 0.088 mAP.

Mechanistic Interpretability Research - Cross-Model Circuit Alignment in Pythia LLMs

Developed a pipeline to transfer attention-head information from Pythia-410M to Pythia-1B for causal circuit localization on the Indirect Object Identification task. Used CKA-based alignment, causal ablations, and statistical evaluation, showing that cross-model alignment recovered comparable behavioral effects to EAP-IG at approximately 22× lower marginal compute, supporting its use as an efficient circuit pre-filter.

TECHNICAL SKILLS

Python: NumPy, SciPy, Pandas, PyTorch, scikit-learn, statsmodels, Seaborn, Matplotlib, Spacy | **R:** Statistics (30/30), Econometrics (28/30) | **SQL:** 30 cum laude/30 Bocconi exam | **LaTeX:** academic experience | **Git:** professional experience

HONORS & AWARDS

- 2024:** MEF Welfare Fund Scholarship (4000€). Merit-based award for 1st and 2nd year university academic results
- 2022:** Register of Excellence, Italian Ministry of Education, Top 3.4% nationally in final high school exams
- 2020:** Olimpiadi di Italiano (Italian Olympiad) - National semifinalist
- 2018:** INPS Programma Itaca Scholarship (€6,000) - Ranked 27th out of 737 applicants nationwide for study abroad program; spent one semester as exchange student in Albuquerque, NM, USA

LANGUAGES

Italian: Native

English: Advanced
(IELTS C1 band 8, 2024)

French: Beginner
(Bocconi course for B1)

Spanish: Beginner